Answer Key - Machine Learning - Undergraduate Level Assessment 21 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) Optimize a convex objective function; B) Can only be trained with stochastic gradient descent; C) Can use a mix of different activation functions; D) None of the above
   Note: Correct option: C - Can use a mix of different activation functions

2. **Answer:** C
   Options: A) 0; B) 1; C) 2; D) 3
   Note: Correct option: C - 2

3. **Answer:** B
   Options: A) 0; B) 1; C) 2; D) 3
   Note: Correct option: B - 1

4. **Answer:** A
   Options: A) P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y); B) P(X, Y, Z) = P(X) * P(Y|X) * P(Z|Y); C) P(X, Y, Z) = P(Z) * P(X|Z) * P(Y|Z); D) P(X, Y, Z) = P(X) * P(Y) * P(Z)
   Note: Correct option: A - P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)

5. **Answer:** A
   Options: A) True, True; B) False, False; C) True, False; D) False, True
   Note: Correct option: A - True, True

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'The polynomial degree'.

8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'they have high variance'.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'True, True'.

10. **Answer:** False
    Options: A) True; B) False
    Note: LLM-generated false statement. Correct answer: 'P(B|A) decreases'.

Long Form Response

11. **Answer:** def check_triplet(A, n, sum, count): | return True | return False | return check_triplet(A, n - 1, sum - A[n - 1], count + 1) or
    Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** def repeat_tuples(test_tup, N): | return (res)
    Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key